

The Psychobiological Effects of Focusing-Oriented Therapy on Heart Rate Variability and Cortisol in Women with PTSD and Complex PTSD: A Pilot Study

Chana Rena Reiser (34681) | cr25790@essex.ac.uk | University of Essex Online | Supervisor: Dr. D. Goldenhaus-Manning

Module: Psychobiology and Neuroscience

Abstract

This pilot tests the psychobiological effects of Focusing-Oriented Therapy (FOT) on 30 women with PTSD/C-PTSD, measuring HRV and cortisol pre- and post-intervention.

Introduction

As a trained Focusing practitioner, this research emerges from direct clinical observation.

Sex differences: Women ~2× more likely to develop PTSD (Tolin & Foa, 2006)

Psychobiology: PTSD linked to ↓ HRV and blunted cortisol (Ge et al., 2020; Metz et al., 2020)

FOT: 89 studies of efficacy but no psychobiological evidence (Krycka & Ikemi, 2016)

Research Question: Does an 8-week FOT intervention change HRV and cortisol in women with PTSD/C-PTSD?

References

Ge, F., Yuan, M., Li, Y., & Zhang, W. (2020). Posttraumatic stress disorder and alterations in resting heart rate variability. *Psychiatry Investigation*, 17(1), 9–20. <https://doi.org/10.30773/pi.2019.0112>

Joshi, S. A., Aupperle, R. L., & Khalsa, S. S. (2023). Interoception in fear learning and posttraumatic stress disorder. *Focus*, 21(3), 266–277. <https://doi.org/10.1176/appi.focus.20230007>

Krycka, K. C., & Ikemi, A. (2016). Focusing-oriented-experiential psychotherapy. In D. J. Cain, K. Keenan, & S. Rubin (Eds.), *Humanistic psychotherapies* (2nd ed., pp. 251–282). American Psychological Association. <https://doi.org/10.1037/14775-009>

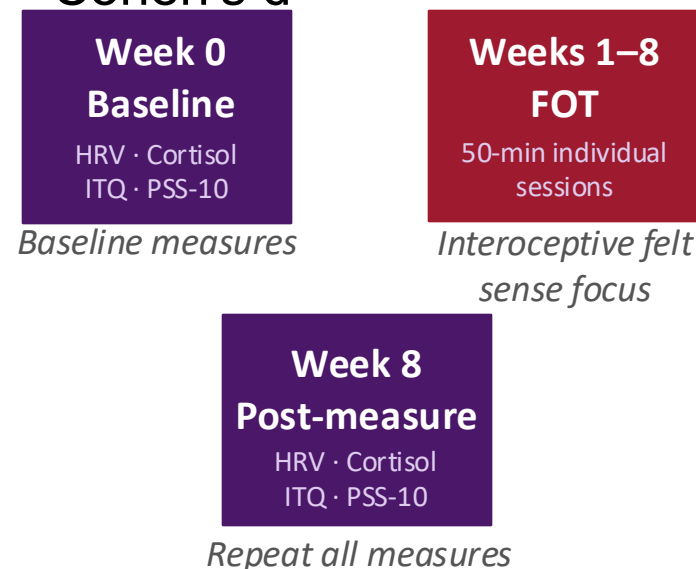
Metz, S., Duesenberg, M., Hellmann-Regen, J., Wolf, O. T., Roepke, S., Otte, C., & Wingenfeld, K. (2020). Blunted salivary cortisol response to psychosocial stress in women with posttraumatic stress disorder. *Journal of Psychiatric Research*, 130, 112–119. <https://doi.org/10.1016/j.jpsychires.2020.07.014>

Porges, S. W. (2021). Cardiac vagal tone. *World Psychiatry*, 20(2), 296–298. <https://doi.org/10.1002/wps.20871>

Tolin, D. F., & Foa, E. B. (2006). Sex differences in trauma and posttraumatic stress disorder. *Psychological Bulletin*, 132(6), 959–992. <https://doi.org/10.1037/0033-2909.132.6.959>

Research Design & Method

- N=30 women; 8-week FOT; pre-post design
- Measures: HRV (ECG) + cortisol (ELISA; 5 time-points × 2 days)
- Analysis: paired t-tests; Cohen's d



Ethics: UoE Ethics approved · Informed consent · right to withdraw · FOT minimizes retraumatization · data anonymized · UK GDPR

Conclusions / Discussion

↑ HRV would support Polyvagal Theory (Porges, 2021)
 Normalized cortisol would support Interoceptive Fear Learning Theory (Joshi et al., 2023)
 If confirmed, this would represent the first psychobiological evidence for FOT in female PTSD · Future RCT warranted

Limitations

No control · small N · pre-post only · women 25–55 only

Take-Home Message:

Focusing-Oriented Therapy may offer a psychobiologically active interoceptive approach for women with PTSD that is theoretically grounded and ethically rigorous.